

PERSONAL FINANCE DASHBOARD - FinSight

Internship Project Report

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ABSTRACT

The Personal Finance Dashboard is a web-based analytics application developed using React, TypeScript, Tailwind CSS, and Recharts. The dashboard enables users to visualize financial information through interactive charts, KPI cards, and transaction analytics.

The application provides a clean and responsive user interface that displays key financial metrics such as income, expenses, savings, and remaining budget. Interactive components including search and category filtering improve usability while charts present financial data in a graphical format for easier analysis.

The project demonstrates the practical application of modern frontend development practices such as component-based architecture, reusable code, responsive design, state management, and data visualization.

INTRODUCTION

Managing personal finance effectively requires users to track their income, expenses, savings, and overall spending habits. Traditional methods such as spreadsheets or handwritten records often make it difficult to analyze financial data and identify spending patterns.

The Personal Finance Dashboard is a web-based application developed to simplify financial monitoring through interactive data visualization and a user-friendly interface. The dashboard provides a centralized platform where users can view important financial metrics using KPI cards, charts, and transaction records.

The project has been developed using modern frontend technologies including React, TypeScript, Tailwind CSS, and Recharts. It demonstrates the implementation of component-based architecture, responsive web design, reusable UI components, and data-driven visualizations.

PROBLEM STATEMENT

Many individuals struggle to monitor and analyse their financial activities due to the absence of simple and interactive financial management tools. Existing solutions are often either too complex for everyday users or require paid subscriptions to access advanced analytics.

The objective of this project is to develop a lightweight, responsive, and visually appealing dashboard that enables users to understand their financial status by presenting income, expenses, savings, and transaction history in an organized and interactive manner.

OBJECTIVES

The primary objectives of the project are:

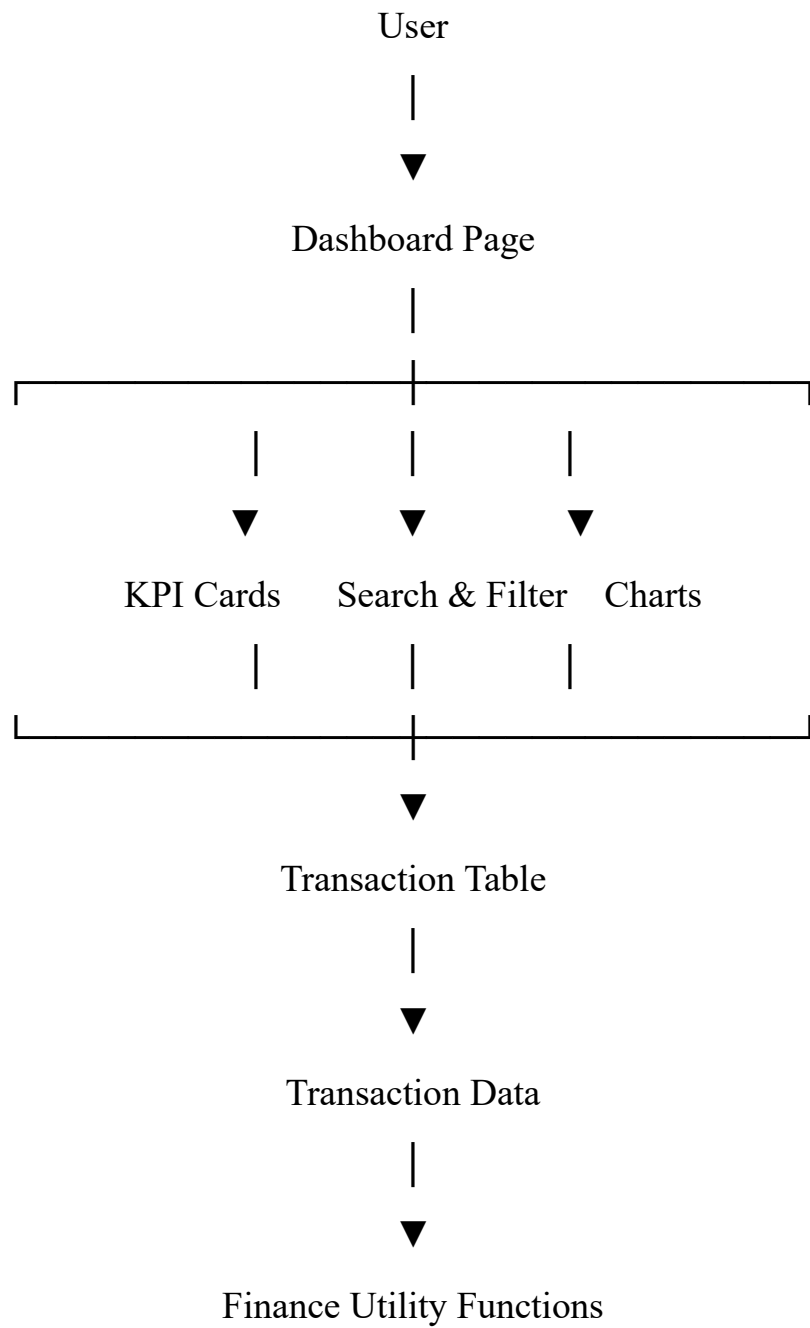
- To design a modern and responsive financial dashboard.
- To display financial summaries using KPI cards.
- To visualize financial data through interactive charts.
- To provide an organized transaction history.
- To implement searching and filtering of transactions.
- To follow a modular and reusable component-based architecture.
- To improve user experience through responsive and intuitive design.
- To strengthen practical knowledge of React and TypeScript development.

TECHNOLOGIES USED

Technology	Purpose
React	Building reusable user interface components
TypeScript	Type-safe JavaScript development
Tailwind CSS	Responsive styling and UI design
Recharts	Data visualization and charts
React Icons	Icon library
Vite	Fast development environment and build tool
HTML5	Page structure
CSS3	Additional styling support

SYSTEM ARCHITECTURE

The application follows a component-based architecture where each UI element performs a specific responsibility.



The Dashboard component acts as the central controller, managing application state and passing data to child components. Utility functions process transaction data, while charts and KPI cards display calculated results.

PROJECT STRUCTURE

src/

|

| — components/

| | — dashboard/

| | | — Charts/

| | | | — IncomeExpenseChart.tsx

| | | | — ExpensePieChart.tsx

| | | | — MonthlyExpenseChart.tsx

|

|

| | — KPICard.tsx

| | — SearchBar.tsx

| | — Filter.tsx

| | | — TransactionTable.tsx

|

|

| | — layout/

| | | — Navbar.tsx

| | | — Sidebar.tsx

| | | — Layout.tsx

|

| — data/

| | — transactions.ts

|

| — pages/

| | — Dashboard.tsx

```
|
|
|— styles/
|   └─ theme.ts
|
|
|— types/
|   └─ transaction.ts
|
|
|— utils/
|   └─ finance.ts
|
|
|— App.tsx
└─ main.tsx
```


IMPLEMENTATION

The dashboard was developed using a modular architecture in which every feature is implemented as an independent React component.

KPI Cards

The dashboard displays four KPI cards showing Total Income, Total Expenses, Total Savings, and Remaining Budget. These values are calculated dynamically from transaction data using reusable utility functions.

Charts

Three different chart types have been implemented using the Recharts library:

- Income vs Expenses Line Chart
- Expense Breakdown Pie Chart
- Monthly Spending Bar Chart

Each chart is generated from transaction data to provide users with graphical insights into their financial activities.

Search Functionality

A controlled search bar allows users to search transactions by description or category. React state management ensures that the displayed transaction list updates instantly as the user types.

Category Filter

Users can filter transactions based on predefined categories, improving accessibility when working with larger datasets.

Transaction Table

The transaction table displays the five most recent financial records with color-coded income and expense values for improved readability.

Responsive Design

Tailwind CSS was used to ensure the dashboard adapts smoothly across desktop, tablet, and mobile devices.

FUTURE SCOPE

The current project serves as the foundation for a complete personal finance management system. Future enhancements may include:

- User authentication and authorization
- Add, edit, and delete transactions
- Date range filtering
- Advanced sorting options
- Budget planning module
- Monthly financial reports
- Export data as PDF or CSV
- Cloud database integration
- REST API integration
- Dark mode support
- AI-powered spending analysis and recommendations

CONCLUSION

The Personal Finance Dashboard successfully demonstrates the practical application of modern frontend development technologies to build a responsive and interactive financial analytics application.

The project applies component-based development, reusable code architecture, state management, and data visualization techniques to provide users with meaningful financial insights. Throughout the development process, emphasis was placed on clean code organization, responsive design, and user experience.

This project strengthened practical skills in React, TypeScript, Tailwind CSS, and data visualization while providing valuable experience in designing scalable frontend applications suitable for real-world scenarios.